529 Project Task 2
SRAM 6T Cell Implementation

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Abstract

This paper illustrates briefly about the operation of 6T SRAM Cell. Six transistors make up the SRAM 6T cell, which is divided into two cross-coupled inverters with access transistors for read and write functions. Positive feedback maintains the cell's stability by guaranteeing that the data contained inside it doesn't change while it is operating normally. We have discussed the gate level leakage and the sub-threshold leakage and what changes need to be made in order to reduce these leakages.

Introduction

Static Random-Access Memory (SRAM), which provides quick and volatile digital information storage, is an essential part of contemporary integrated circuits. A key component of SRAM arrays, the SRAM 6T cell is frequently used in cache memories and other memory architectures. An outline of the SRAM 6T cell's functionality is provided in this abstract, with particular attention paid to the cell's read/write stability and important design factors.

Six transistors make up the SRAM 6T cell, which is divided into two cross-coupled inverters with access transistors for read and write functions. Positive feedback maintains the cell's stability by guaranteeing that the data contained inside it doesn't change while it is operating normally. Controlling the access transistors during read and write operations allows for the storage and retrieval of data.

By detecting the voltage differential between the complementary bitlines and using that information to drive one of the inverters to dominate and reinforce the data that has been stored, read stability is preserved. Write operations allow data to be written into the cell by modifying the states of the cross-coupled inverters through bitline and wordline control. A major design problem for SRAM cells is balancing read and write stability, speed, and power consumption.

The SRAM 6T cell's design takes into account factors including transistor sizing, minimising leakage current, and utilising cutting-edge process technologies. Low-threshold voltage transistors, adaptive body biassing, and help circuits are some of the strategies used to maximise efficiency while resolving power and variability issues.

Planning a 6T SRAM cell with perused and type in circuitry includes tending to different building challenges to attain a adjust between execution, control utilization, zone productivity, and soundness. Underneath are a few of the key challenges confronted amid the development of a 6T SRAM cell:

It is essential to comprehend how the SRAM 6T cell works when creating dependable and effective memory systems. This abstract illustrates the difficulties and design optimisations associated with the SRAM 6T cell and offers insights into the fundamental ideas controlling its behaviour. The construction of the 6T SRAM cell with read and write circuitry presented issues in user engineering and design.

I have faced the below challenges when designing the circuitry of 6T SRAM, In order to balance performance, power consumption, space efficiency, and stability, a number of engineering issues must be addressed during the design of a 6T SRAM cell with read and write circuitry. The following are some of the main difficulties encountered when building a 6T SRAM cell:

Read vs Write Trade - off: One of the biggest challenges is finding a balance between write margin and read stability. The write margin may be decreased by increasing read stability,

which would affect the dependability of write operations. To maximise the trade-off, apply sophisticated circuit approaches including dynamic voltage scaling, write assist circuits, and read assist circuits.

Power Usage and Leakage: Leakage currents can greatly increase power consumption, particularly in sub-threshold areas, and SRAM cells are known for their high power consumption. Use adaptive body biasing, sleep transistors, and power gating to cut down on leakage currents during standby times. Use low-threshold voltage transistors and low-power design strategies.

Density and Size of Cells: Obtaining a small cell size without sacrificing stability and read/write margins is difficult since the 6T architecture calls for numerous transistors. Investigate multi-port and multi-bit cell layouts to increase density and use cutting-edge process technologies to shrink transistor sizes.

Variability and Sensitivity of Processes: SRAM cells exhibit variability in their characteristics due to their sensitivity to process modifications. To lessen the effects of process changes, use redundancy approaches and assist circuits. Apply statistical design techniques to take variability into consideration.

Write Disturb and Half-Select Disturb: Half-select disturb occurs when a cell is only partially selected during a read operation, while write disturb occurs when writing to one cell impacts other cells. To reduce the impact of write and half-select disturbances, apply sensing techniques and error correcting codes (ECC). Take great care when creating the read and write pathways to reduce disruptions.

Velocity and Time of Access: It can be difficult to achieve high-speed read and write operations without sacrificing stability. To increase speed without sacrificing stability, optimise transistor sizes, use read and write assist strategies, and investigate pipeline and parallel access architectures.

Transition of Process Technology Node: As technology nodes get smaller, new problems including lower noise margins and higher variability appear. To adapt to smaller technology nodes, use cutting-edge process-aware design techniques and investigate new materials and transistor architectures.

Sensitivity to Temperature: SRAM cell performance and stability can be impacted by temperature changes. Use thermal sensors and temperature compensation techniques to dynamically modify operating parameters in response to temperature changes.

To tackle these obstacles, a multidisciplinary strategy incorporating advancements in process technology, device physics, and circuit design is necessary. It will take ongoing research and development to overcome these obstacles and advance SRAM technology.

FinFET Transistors:

As features get smaller, a FinFET (Fin Field-Effect Transistor) can overcome the drawbacks of conventional CMOS (Complementary Metal Oxide Semiconductor) transistors. FinFETs are a significant improvement in semiconductor technology. The earliest 2000s saw the development of FinFETs, a semiconductor technology that emerged as a result of researchers and manufacturers searching for novel transistor designs to address the difficulties associated with miniaturisation.

Researchers experimented with three-dimensional transistor architectures to get improved power consumption, energy economy, and performance for short-channel control. FinFETs, which have shown to be a workable solution for this issue, offer improved electrostatic control over the channel thanks to their distinctive finlike structure. Fin-shaped transistors were first conceptualised in the late 1990s, although they weren't widely used or put into practice until after 2000.

In 2011 the initial integration of FinFETs into commercial processors were introducted by Intel, which was a major turning point for the semiconductor sector. Later, other significant semiconductor manufacturers like Samsung and TSMC (Taiwan Semiconductor Manufacturing Company) incorporated FinFET technology into their manufacturing processes.

The main factor driving the popularity of FinFETs is their capacity to overcome the difficulties posed by planar CMOS technology as transistor sizes drew closer to the nanoscale. FinFETs use a vertical fin structure that extends from the silicon substrate to assist current flow management. This design not only reduces leakage current and reduces short-channel effects, but it also makes it possible to expand transistor size beyond what was previously achievable with planar transistors. FinFETs are crucial to the design of digital circuits in both Standard Domino logic and Dual-Vt Domino logic, as they offer unique benefits in meeting particular design specifications.

The basic architecture of a FinFET is distinct from that of a conventional planar CMOS (Complementary Metal-Oxide Semiconductor) transistor. FinFETs have a fin-like structure that rises vertically from the silicon substrate in place of a flat channel. The transistor's electrostatic control is improved by this fin structure, which offers several surfaces for controlling current flow. It also lessens the negative effects of scaling, such as increased leakage current and short-channel effects.

The three-dimensional geometry of finFETs increases performance and lowers power consumption by providing superior gate control over the channel compared to planar transistors. This kind of structural innovation has been crucial to maintaining semiconductor manufacturing along the path indicated by Moore's Law.

Influence on Energy Efficiency: The improvement in power efficiency is the primary benefit of FinFET design. As the semiconductor industry moves towards smaller process nodes,

where power dissipation becomes a major concern, this enhancement becomes even more important.

Decreased Leakage Current: One of the main elements influencing power consumption in conventional planar transistors is leakage current, which FinFETs naturally control. When the transistor is in the off state, the vertical fin construction reduces leakage and provides greater control over the flow of current.

Reduced Operation Voltage: Compared to planar semiconductors, finFETs can function at lower supply voltages. This lower working voltage is consistent with the increased emphasis on low-power designs across a range of applications, including Internet of Things (IoT) devices and mobile devices, and it helps to reduce overall power consumption.

Enhancing Device Performance: Because FinFET architecture allows for increases in speed and total processing power, it is essential for enhancing device performance beyond power savings.

Controllability of Short-Channel Effects: FinFETs' fin-like shape gives them more control over short-channel effects as transistors get smaller. Increased scaling of transistor size is possible with this control without compromising performance.

Higher Rates of Switching: The three-dimensional fin structure's improved electrostatic control makes it possible to switch at a faster rate. FinFETs facilitate quicker transitions between the on and off states, which enhances circuit performance.

Beyond Linear Limitations Expansion: Compared to planar transistors, FinFETs have greatly increased the scalability of semiconductor devices. This scaling allows for the integration of complicated activities and increases processing capacity by cramming more transistors onto a chip.

Factors related to process and manufacturability: The manufacturing process issues and implementation hurdles in semiconductor manufacture are included in the architectural analysis of FinFETs.

Complex Manufacturing Process: Compared to typical planar transistors, the manufacture of FinFETs involves a more intricate process. The fin-like features complicate the process and raise the possibility of yield and manufacturing cost issues because they require extra fabrication stages.

Process Variability Management: FinFETs present extra issues because to their fin size and how it affects device properties, even while they enable enhanced control over some types of process variations. For a variety of manufactured devices to operate consistently, process variability control becomes essential.

Architectural Exploration in Standard Logic and Specialized Designs:

Investigating how FinFET technology impacts various semiconductor designs—from generic logic gates to tailored circuits for specific applications—is necessary for the deployment of the technology.

Typical Logic Gates: Conventional logic gates like NAND and NOR gates have been modified to incorporate FinFETs in order to leverage their advantages in terms of performance and power efficiency. As part of the architectural exploration process, FinFET placement within these gates is tuned to obtain the desired trade-off between speed and power.

Expert Designs (such as Dual-Vt Domino Logic): FinFETs are key components in specialised circuit designs such as Dual-Vt Domino Logic, where transistors with two different threshold voltages are intentionally used. FinFETs' three-dimensional nature facilitates the integration of dual-threshold transistors more easily, allowing designers greater latitude.

Challenges and Prospective Paths: Although FinFET technology has led to advancements in semiconductor design, architectural research still needs to anticipate future directions and overcome challenges.

Scope of discoveries: Increased design complexity: The three-dimensional structure of FinFETs adds another degree of intricacy to design, particularly in terms of managing transistor placement and dual-threshold devices. For these in order to effectively handle complexity, design tools and approaches must progress.

Heat Transmission: As devices develop smaller and more powerful, it becomes more important to manage heat dissipation. To prevent heat issues that can jeopardise reliability and performance, architectural Effective heat dissipation techniques need to be taken into account during investigation.

Architectural Exploration in Standard Logic and Specialized Designs:Persistent Scaling Difficulties: FinFETs have enabled scaling beyond planar bounds, but further advancement will probably be hampered by quantum effects and physical limitations. Novel technologies, such as nanosheet transistors, are being created.

Standard Domino Logic with FinFET: To perform logic operations, the standard Domino logic dynamic logic family precharges and discharges nodes. FinFETs have a number of advantages when applied to Standard Domino logic.

Improved Control over Short-Channel Effects: The three-dimensional structure of FinFETs inherently provides more control over short-channel effects, hence improving the performance and dependability of dynamic logic operations.

Reduced Current Leakage: FinFETs enable the reduction of leakage current, especially in dynamic logic applications where nodes need to be precharged and discharged. This decrease in leakage leads to an increase in power efficiency.

Better Performance and Speed: Because of their improved electrostatic control, FinFETs perform better overall and in dynamic logic circuits at quicker switching speeds.

FinFET in Dual-Vt Domino Logic: To maximise efficiency and minimise power consumption, dual-Vt Domino logic uses transistors with two different threshold voltages (Vt). When used with FinFET technology, Dual-Vt Domino logic offers the following additional benefits:

Tailored Threshold Voltage: By incorporating transistors with two threshold voltages, FinFETs enable circuit designers to customise the threshold voltages for various circuit sections. In non-critical channels, high-Vt transistors can minimise leakage whereas low-Vt transistors can boost speed.

Optimising through Strategic Placement: FinFETs' three-dimensional structure facilitates the strategic placement of high- and low-Vt transistors in critical and non-critical channels. This optimisation helps to achieve the desired ratio of speed to leakage power.

Improved Process Variability Handling: Due to FinFETs' improved control over short-channel effects, Dual-Vt systems are able to handle process fluctuations more skillfully. Maintaining performance consistency in a variety of manufacturing scenarios requires this.

Power Dissipation: Although FinFETs in Dual-Vt architectures have the ability to intentionally reduce leakage power, it is still necessary to consider the dynamic power dissipation that occurs throughout each clock cycle. Careful balance is necessary to fully benefit from Dual-Vt without compromising overall power efficiency.

With their three-dimensional fin-like shape, FinFETs have become a major invention in semiconductor manufacturing today, helping to overcome the difficulties brought on by the shrinking dimensions of transistors. To name a few, the basic architecture of FinFETs and its effects on performance, power efficiency, and general design considerations.

In conclusion, the examination of FinFET technology from an architectural standpoint represents a significant advancement in the field of semiconductor history. The three-dimensional fin-like structure has revolutionized transistor design by providing advantages in power efficiency, performance, and scalability. As the industry pushes the boundaries of miniaturization, the next generation of high-performance, energy-efficient semiconductor devices will be realized through the investigation of FinFETs and the integration of innovative architectural solutions.

Multi-Threshold CMOS (MTCMOS):

The goal of effective power management in contemporary integrated circuits is what gave rise to the creation of Multi-Threshold CMOS (MTCMOS) technology. Power consumption became a major concern with the advancement of semiconductor technologies, especially in battery-powered gadgets and energy-efficient computing systems. The origins of MTCMOS can be found in the late 20th century, when engineers and scientists were looking for novel ways to reduce power dissipation in complementary metal-oxide-semiconductor (CMOS) circuits.

Traditional power gating methods first appeared as a means of lowering power consumption during idle times as CMOS technology developed. Nevertheless, issues with leakage currents and the growing importance of sub-threshold leakage in deep sub-micron technologies presented difficulties for these methods. The goal of MTCMOS technology is to improve integrated circuits' overall energy efficiency by addressing these issues and offering a more sophisticated power management solution.

Architectural Importance: Using transistors with various threshold voltages in a single circuit is what defines MTCMOS. The concept of multiple threshold voltages is introduced by MTCMOS, which allows for better control over the power consumption of various circuit portions. Traditional CMOS transistors have a fixed threshold voltage. The capacity of MTCMOS to dynamically modify threshold is what gives it architectural significance.

Power gating in MTCMOS architectures is accomplished by selectively cutting off power to circuit elements that are not in use during idle times. Leakage currents can be greatly decreased in the power-gating network by using transistors with higher threshold voltages. Performance may be maintained while still achieving efficient power management because to MTCMOS's adjustable threshold voltages.

Advantages of MTCMOS:

Dynamic Power Management: By allowing threshold voltages to be dynamically adjusted, MTCMOS offers efficient power management that is dependent on the operational needs of various circuit blocks. This flexibility reduces the amount of static power used in both active and idle conditions.

Reduction of Leakage Current: In power-gating networks, the use of transistors with higher threshold voltages reduces sub-threshold leakage currents during standby times. This is especially important for deep sub-micron technology, since leakage eventually occurs there is power dissipation too.

Increased Energy Efficiency: MTCMOS helps to increase energy efficiency by selectively shutting down inactive parts of a circuit, which makes it a good fit for battery-powered gadgets and energy-conscious applications.

Improved Performance-Per-Watt: Modern computer systems require a greater performance-per-watt ratio, which is made possible by MTCMOS's capacity to optimise power usage without sacrificing performance.

Adaptability to Technology Scaling: As semiconductor technologies continue to shrink, MTCMOS technology is still relevant. Because of its capacity to adjust to fewer technology nodes, it is a useful remedy for advanced integrated circuit power issues.

Disadvantages of MTCMOS:

Complex Design: Because MTCMOS incorporates multiple threshold voltages, it complicates circuit design. The trade-offs between complexity and the intended power-saving advantages must be carefully considered by designers.

Increased Design Work: Meticulous design work, encompassing transistor sizing, control circuitry, and power-gating techniques, is necessary to achieve optimal power reductions using MTCMOS. Costs and development times may be impacted by this increased design work.

Possible Performance Impact: There is a chance that the overhead of the threshold voltage switching and control circuitry will have an effect on performance in specific situations. In MTCMOS design, striking a balance between power savings and performance requirements is essential.

Technology-Dependent: The particular semiconductor technology and manufacturing method used determine how effective MTCMOS is. It might not benefit every technological node to the same extent.

MTCMOS Use in the Read and Write Circuitry of the SRAM 6T Cell:

A key component of static random-access memory (SRAM) and a key component of high-speed data storage is the SRAM 6T cell. The read and write circuitry of SRAM 6T cells uses MTCMOS technology to overcome power consumption issues that come with conventional SRAM designs.

Read circuitry:

The bitline circuitry and sensing amplifier in an SRAM 6T cell read operation can be implemented using MTCMOS technology. Through dynamic threshold voltage adjustments made during the read process, MTCMOS lowers sense amplifier power consumption without sacrificing read stability. This flexibility minimises static power dissipation during read cycles by ensuring that only the components that are required are powered.

Write Circuitry:

By maximizing the power consumption of the word line and bitline circuitry, MTCMOS helps with the write process. Leakage currents and total power dissipation can be decreased during a write operation by selectively powering the write-related components thanks to MTCMOS technology. This is especially significant in situations where a high write throughput is required and power efficiency is critical.

Voltages at Adaptive Threshold:

The SRAM 6T cell may adaptively set the threshold voltages of the transistors involved in the read and write channels thanks to the usage of MTCMOS. For example, lower threshold voltages are used to preserve performance during active read and write operations, whereas higher threshold voltages can be used to minimize leakage currents during idle times.

Selective Power Gating:

MTCMOS enables dynamic control of power consumption based on operating requirements by enabling selective power gating of certain sections of the SRAM 6T cell. This flexibility maximizes energy efficiency by ensuring that only the components that are required are energized.

Balancing Performance and Power:

The trade-offs between performance and power savings must be carefully considered when integrating MTCMOS into SRAM 6T cells. For designers to obtain the desired power, a balance must be struck.

Functional Validation and Verification:

Verifying that the 6T SRAM cell operates as planned within the given operating limits is known as functional validation. This procedure usually consists of a number of simulations and experiments to confirm how the cell behaves in various situations.

The 6T SRAM cell should precisely output the stored data during a read operation without modifying the stored information. To guarantee accurate data retrieval, functional validation entails replicating read processes under multiple circumstances, such as distinct data patterns, read delays, and outside noise.

The 6T SRAM cell is used to store fresh data during the write operation. Functional validation makes assurance that the cell can write data consistently and without interfering with nearby cells. Write operation tests include different operating conditions, write delays, and input data.

Validation of Write Operation: In the write operation, fresh data is stored in the 6T SRAM cell. Functional validation makes assurance that the cell can write data consistently and

without interfering with nearby cells. To verify the stability and integrity of the stored data, write operation tests are conducted with different input data, write delays, and operating conditions.

Stability Checks: SRAM cells' stability is a vital component. Tests that evaluate the stability of the stored data during read and write operations are part of functional validation. Evaluation of the cell's behaviour over time, at various temperatures, and in the presence of process changes is the goal of stability checks.

Power Consumption Analysis: A crucial component of functional validation is determining the 6T SRAM cell's power consumption. Simulating a cell's behaviour in various power modes, such as active, standby, and sleep, is known as power analysis. This guarantees the cell satisfies power consumption requirements and helps the system run more energy-efficiently.

Testing for Noise and Sensitivity: External disturbances and noise might affect the 6T SRAM cell. Tests to determine the cell's sensitivity to noise, electromagnetic interference, and other outside variables are part of the functional validation process. This guarantees the cell's ability to function dependably in realistic, everyday settings.

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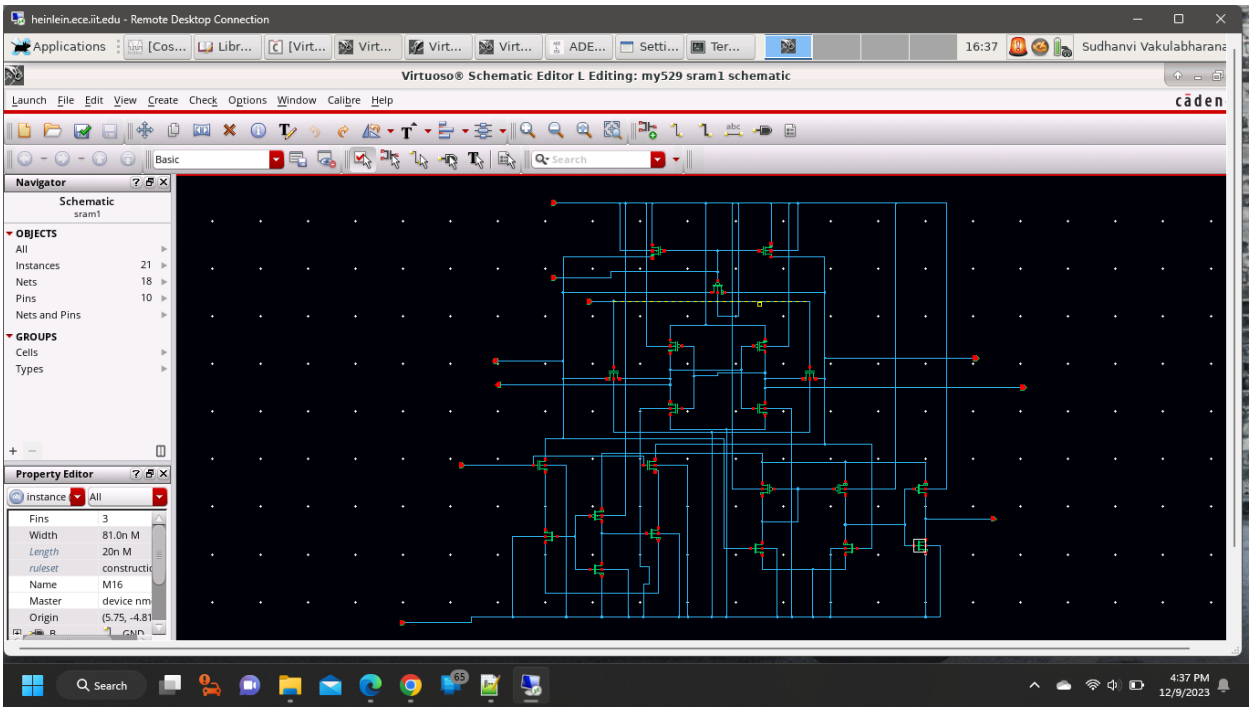
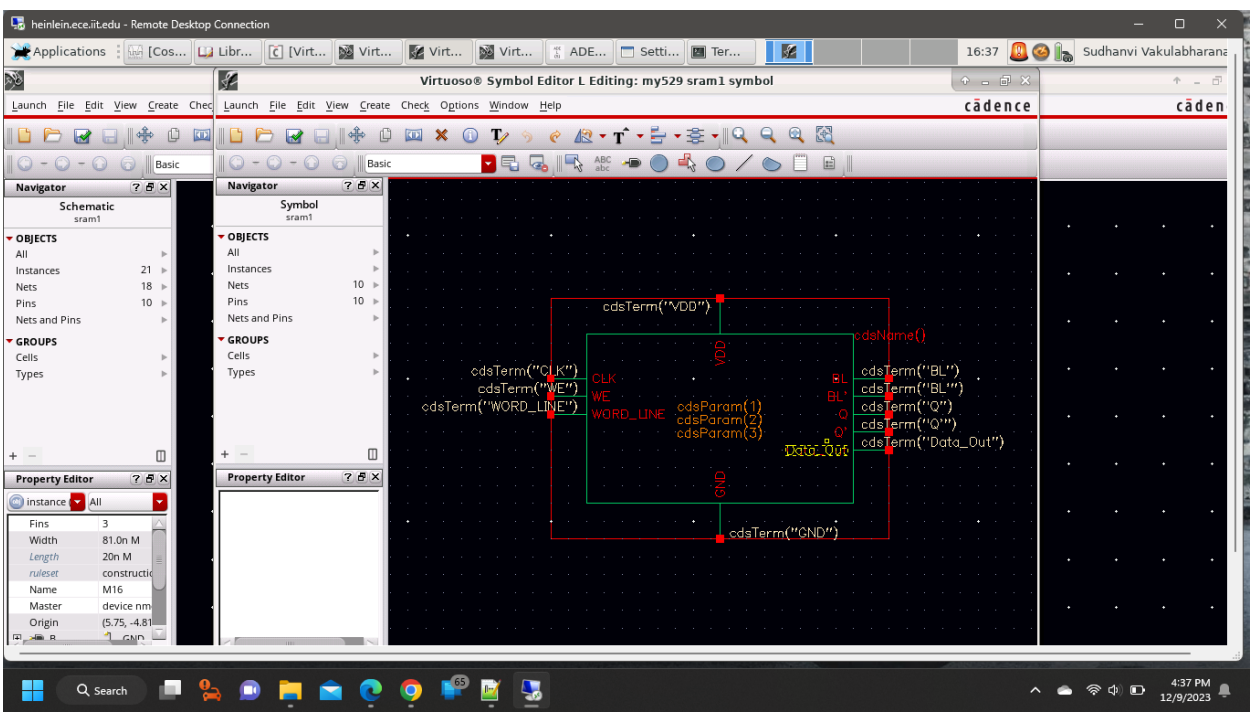
State	Read 0	Read 1	Write 0 (Cell holding 0)	Write 1 (Cell holding 0)	Write 0 (Cell holding 1)	Write 1 (Cell holding 1)
0	0	1	0	1	0	1
1	0	1	0	0	1	1

State table for sram cell

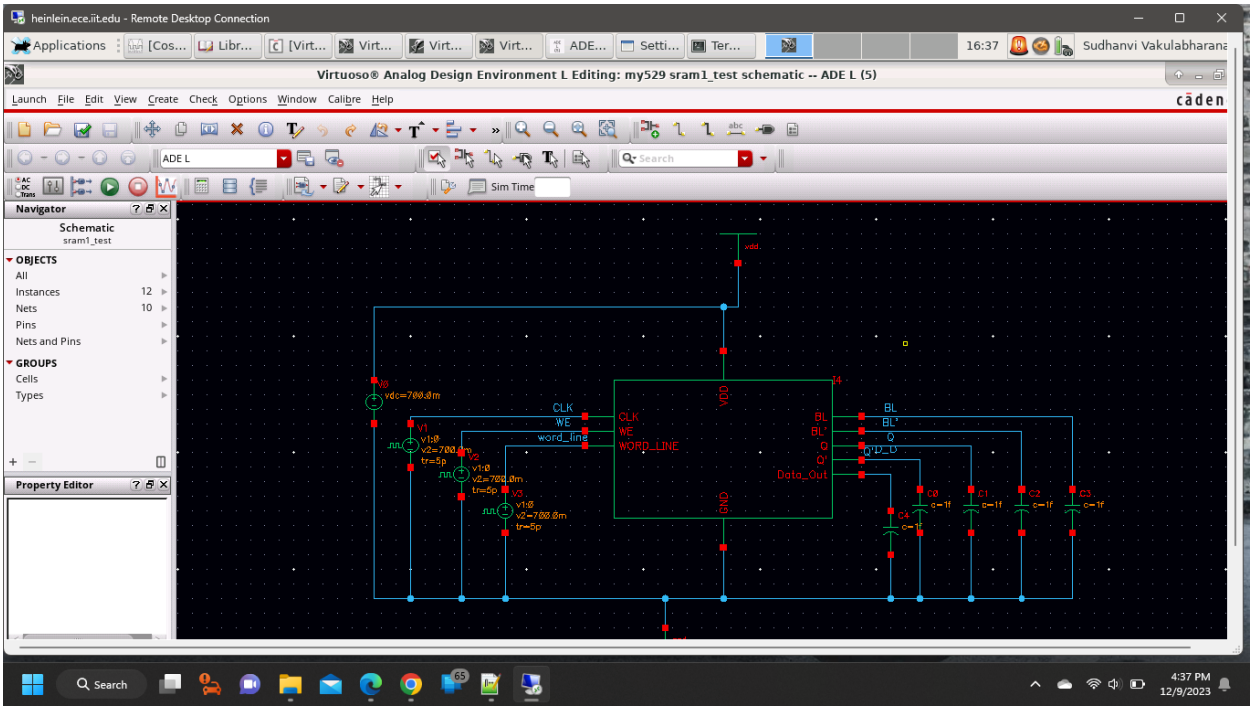
Results:



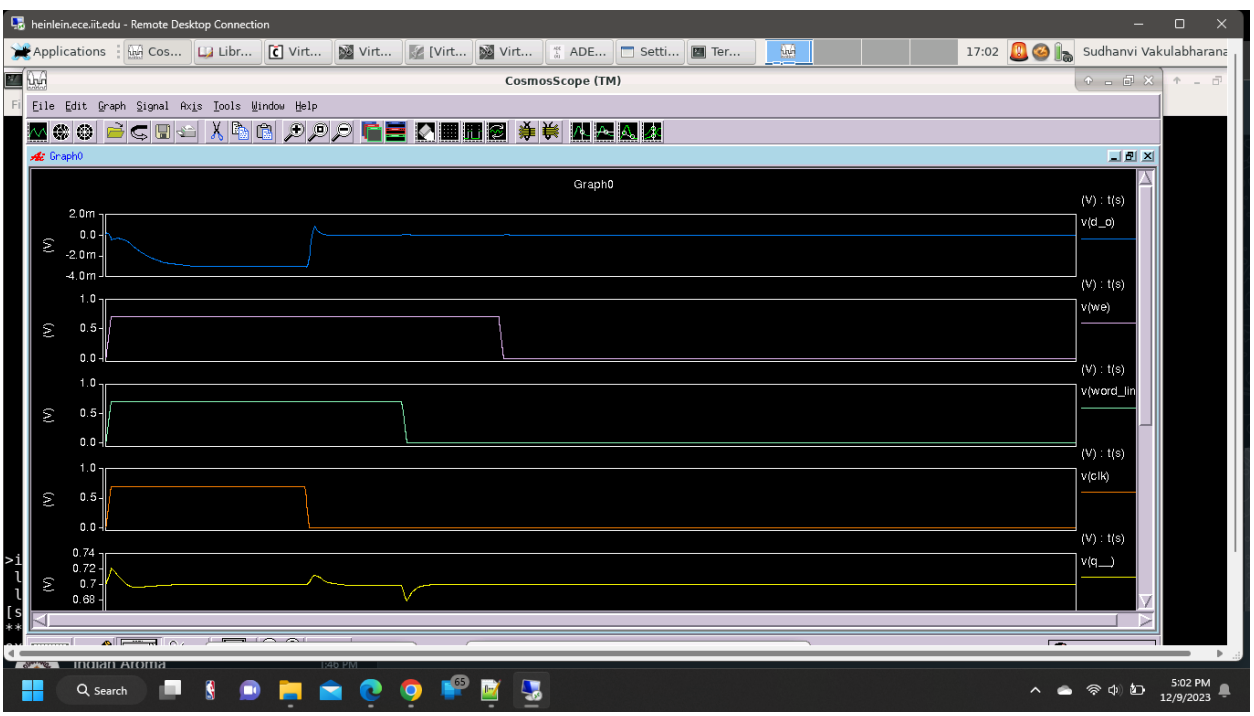
Schematic of Normal 6T SRAM cell

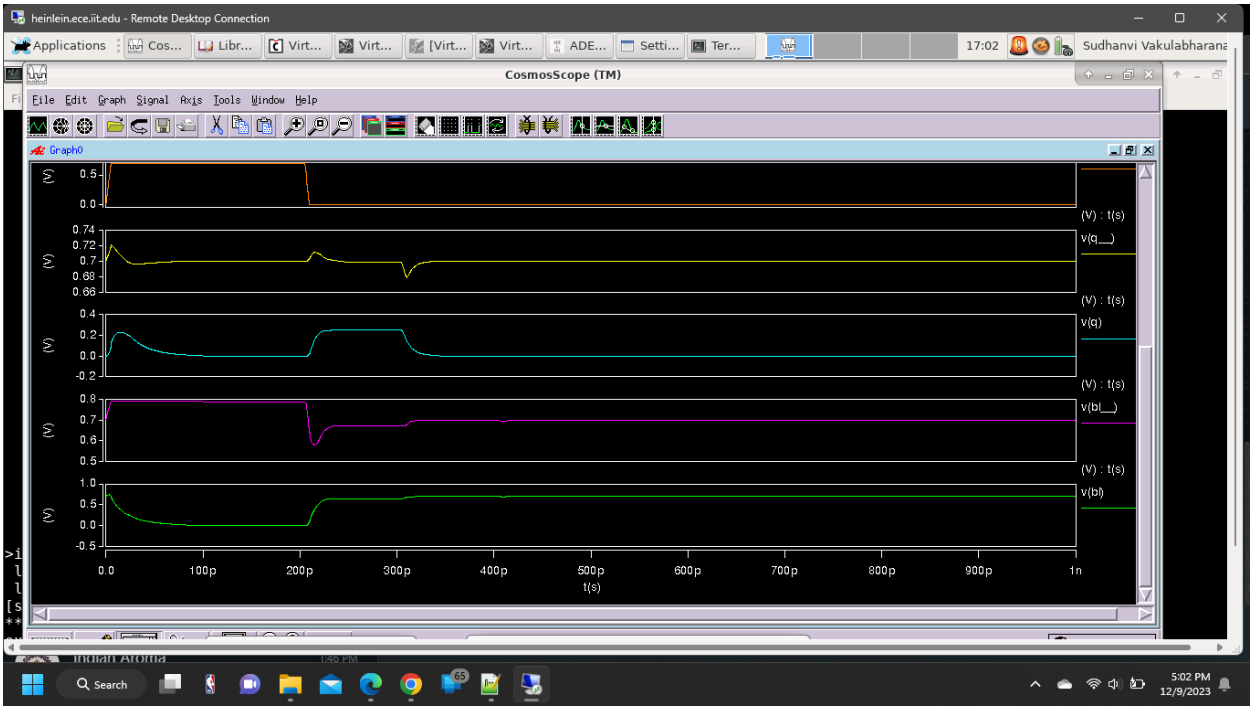


Symbol for Normal 6T SRAM Cell

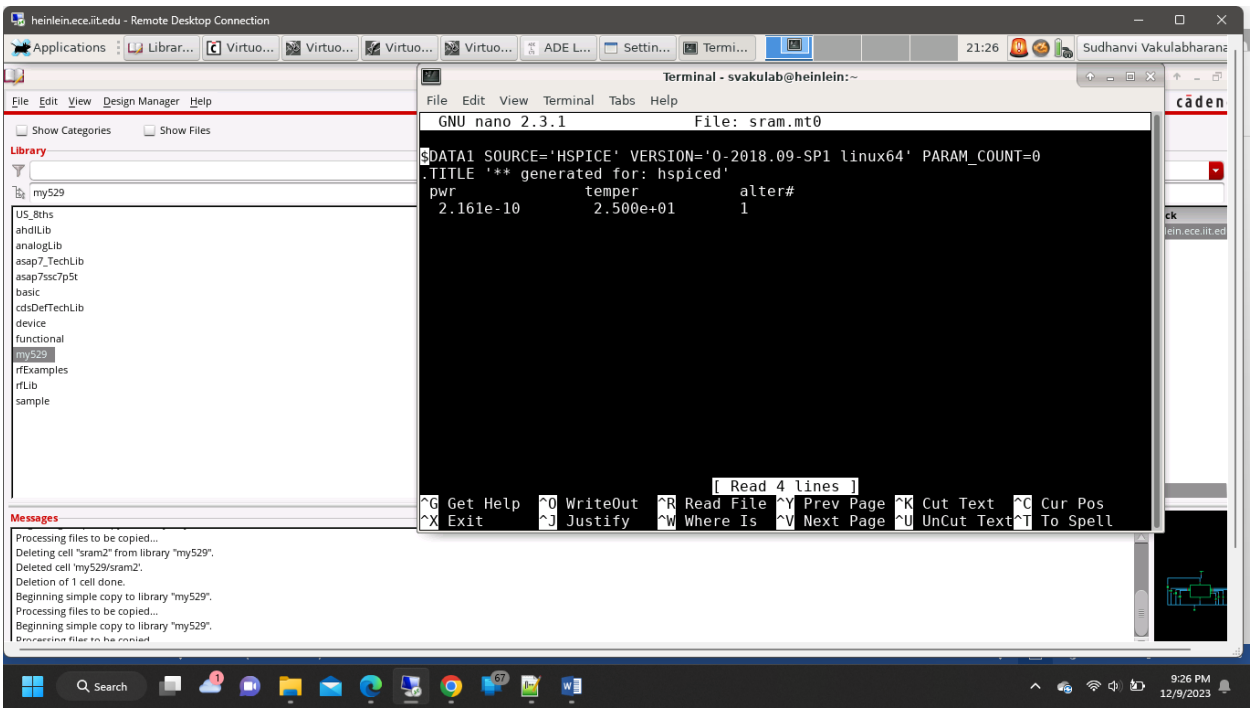


Test Schematic for Normal 6T Cell

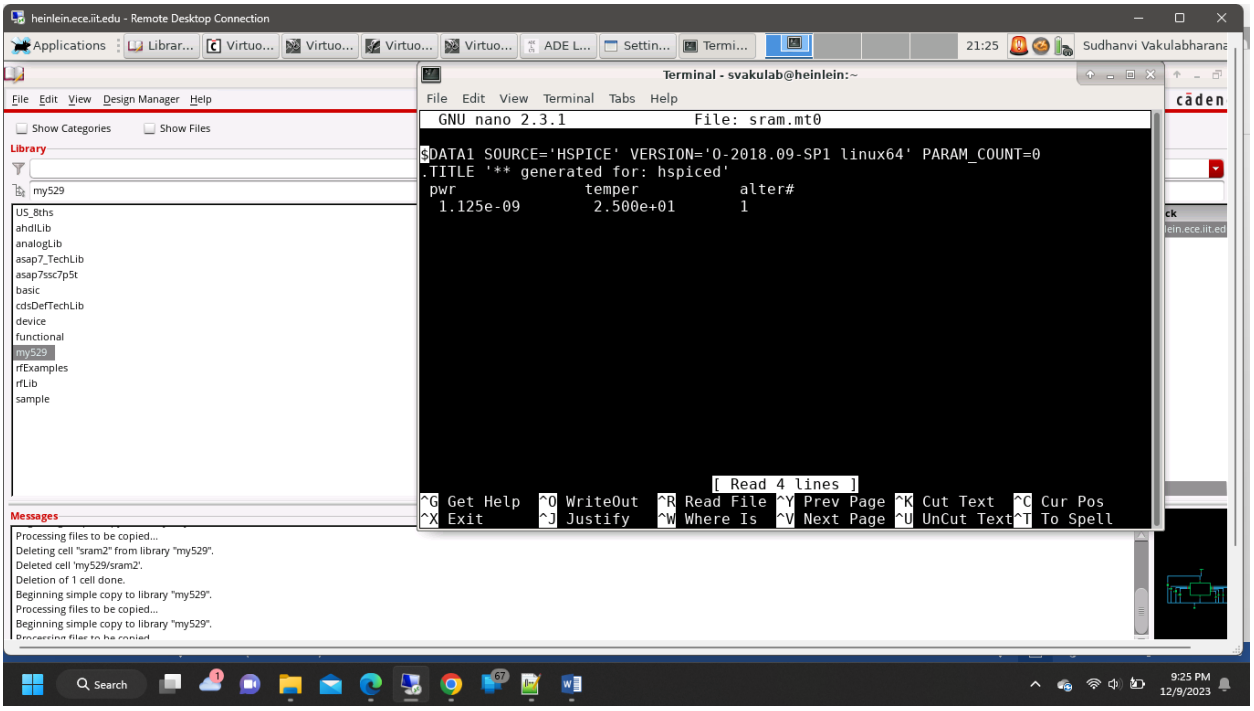




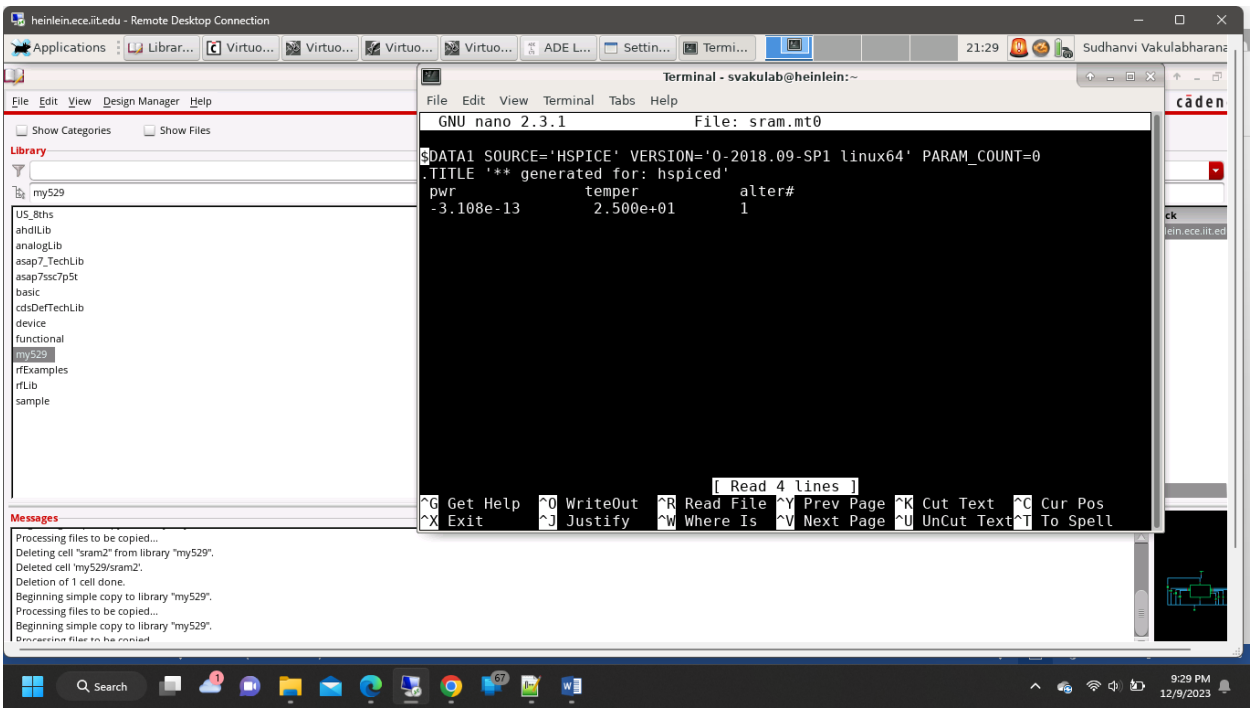
Graphs for Normal 6T SRAM Cell



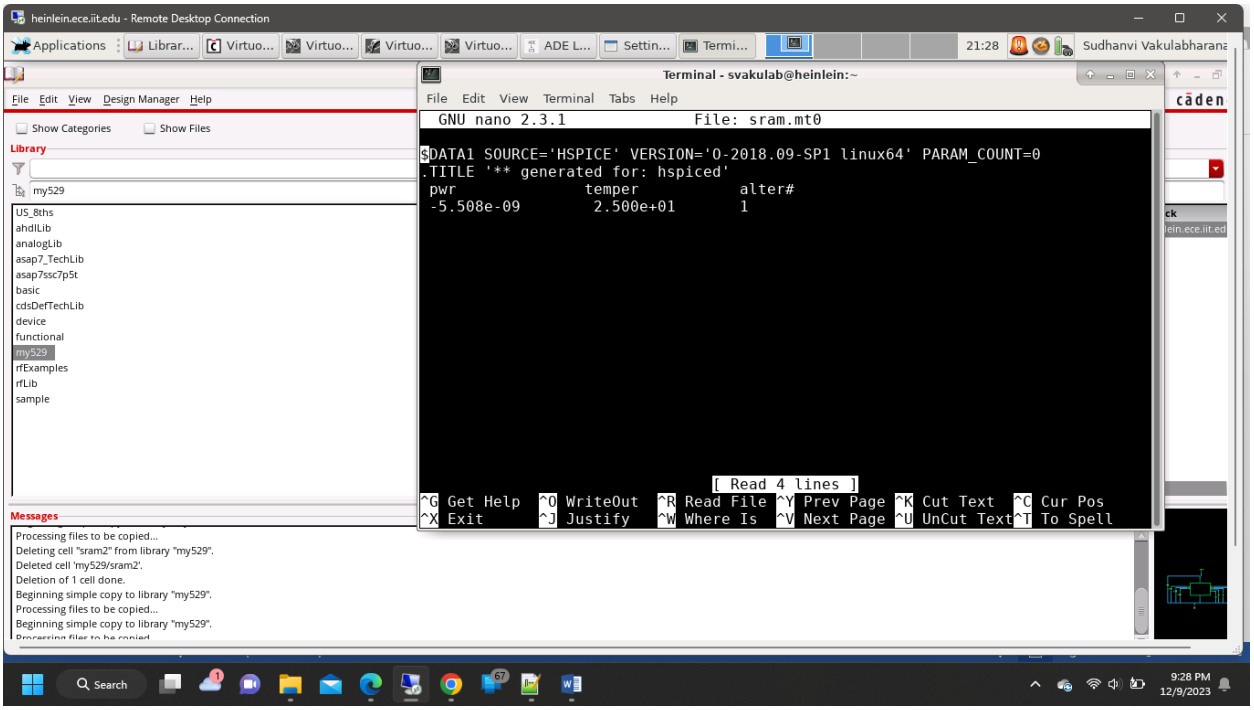
Dynamic Power at Capacitor C0



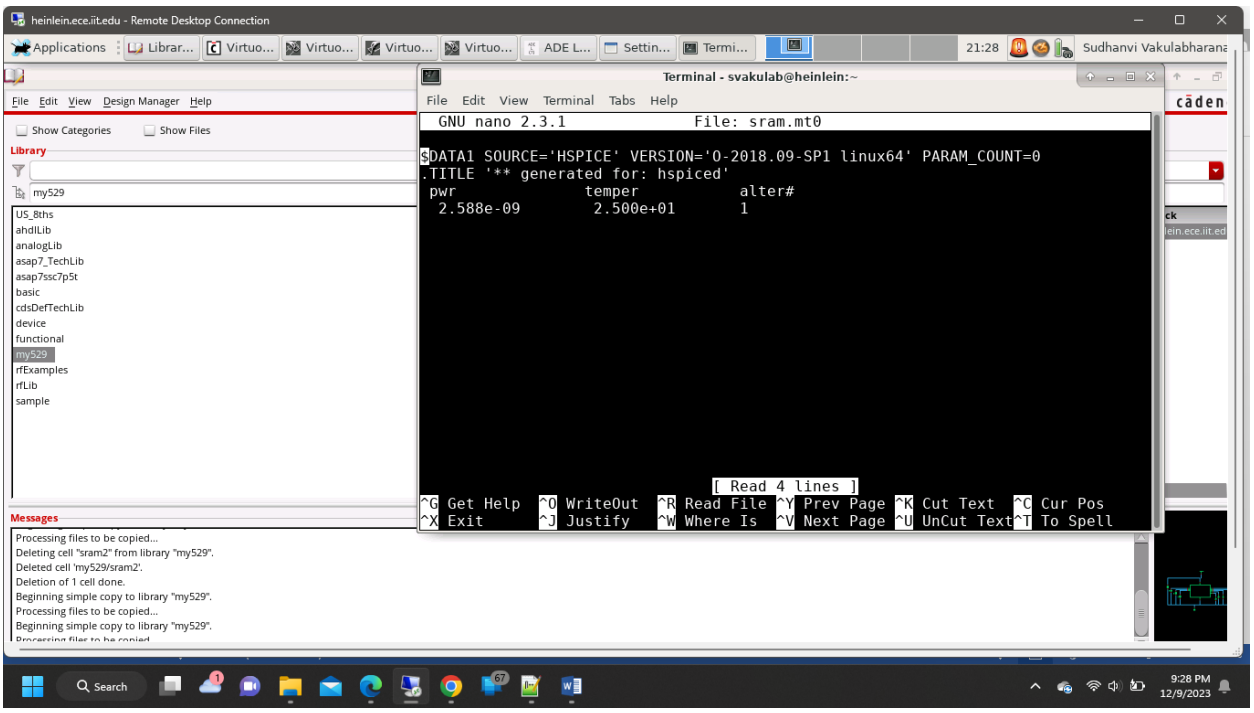
Dynamic Power at Capacitor C1



Dynamic Power at Capacitor C2

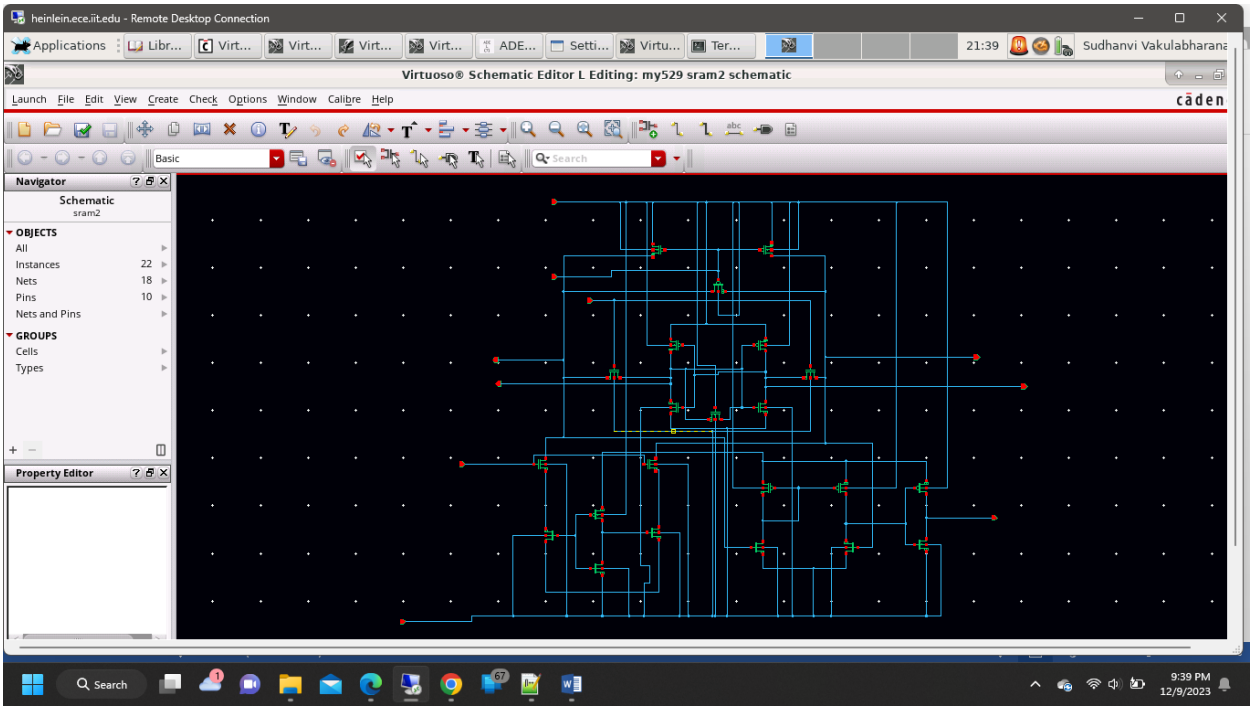


Dynamic Power at Capacitor C3

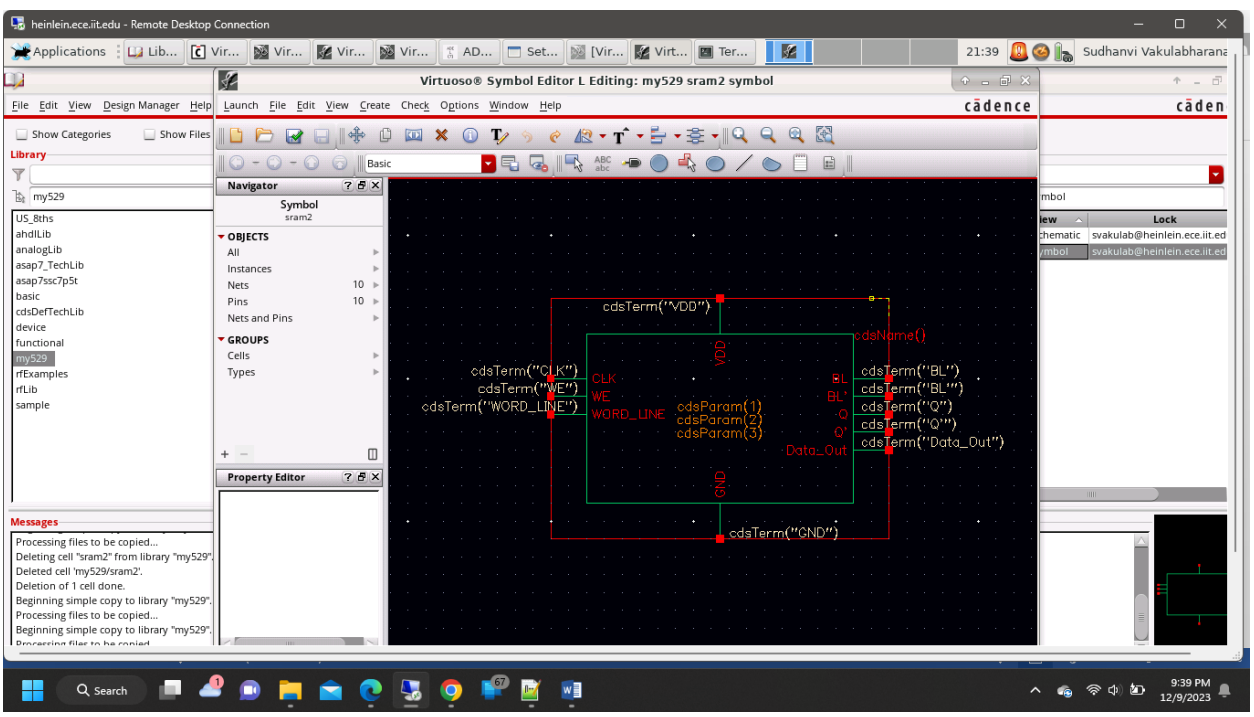


Dynamic Power at Capacitor C4

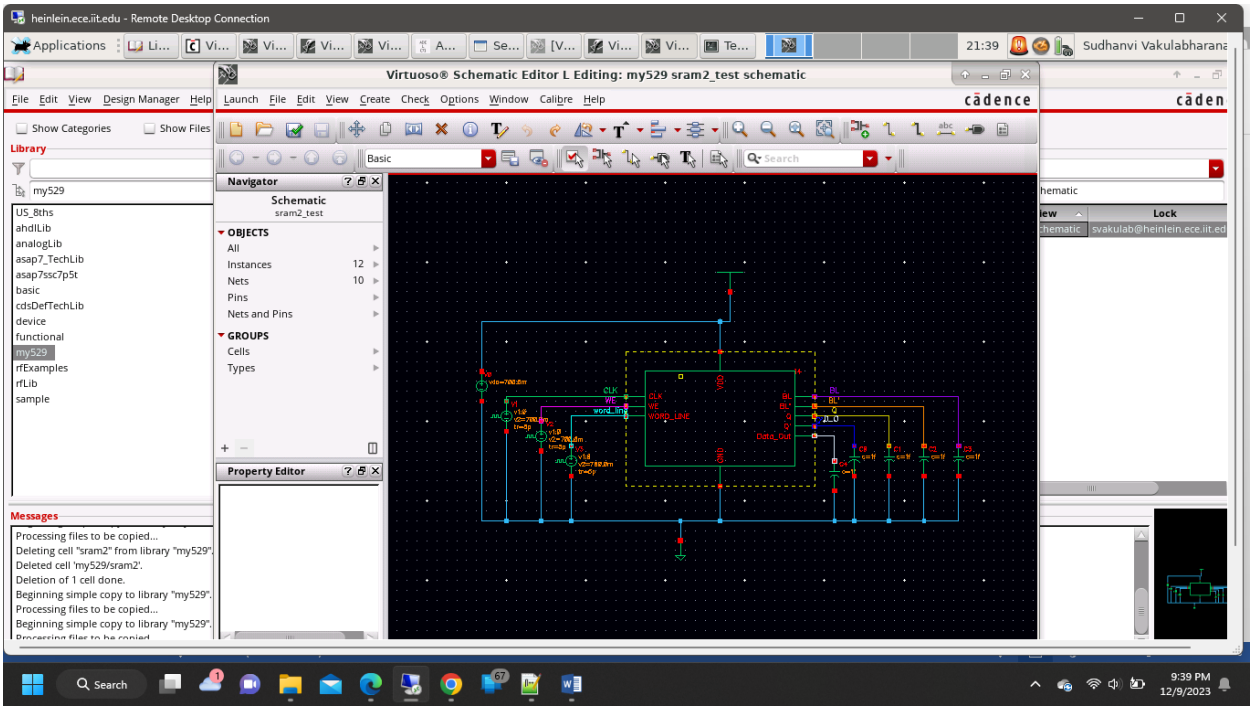
SRAM 6T Optimised:



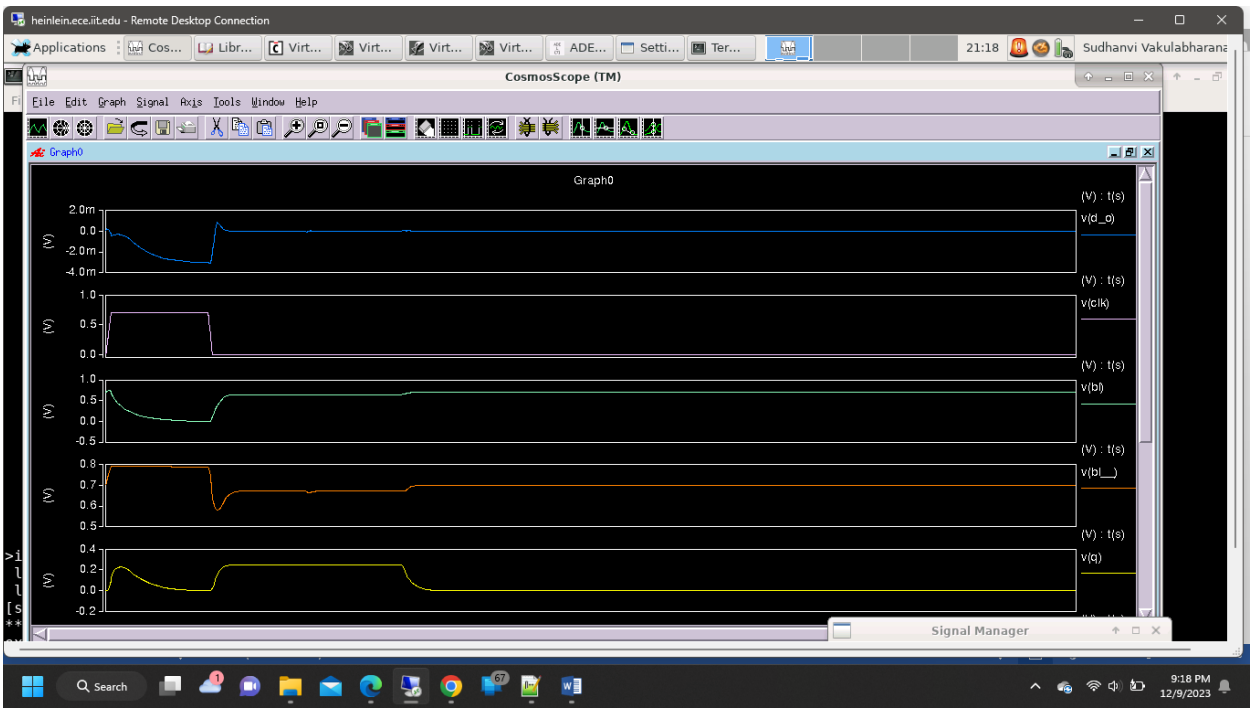
Schematic of Optimised 6T SRAM Cell

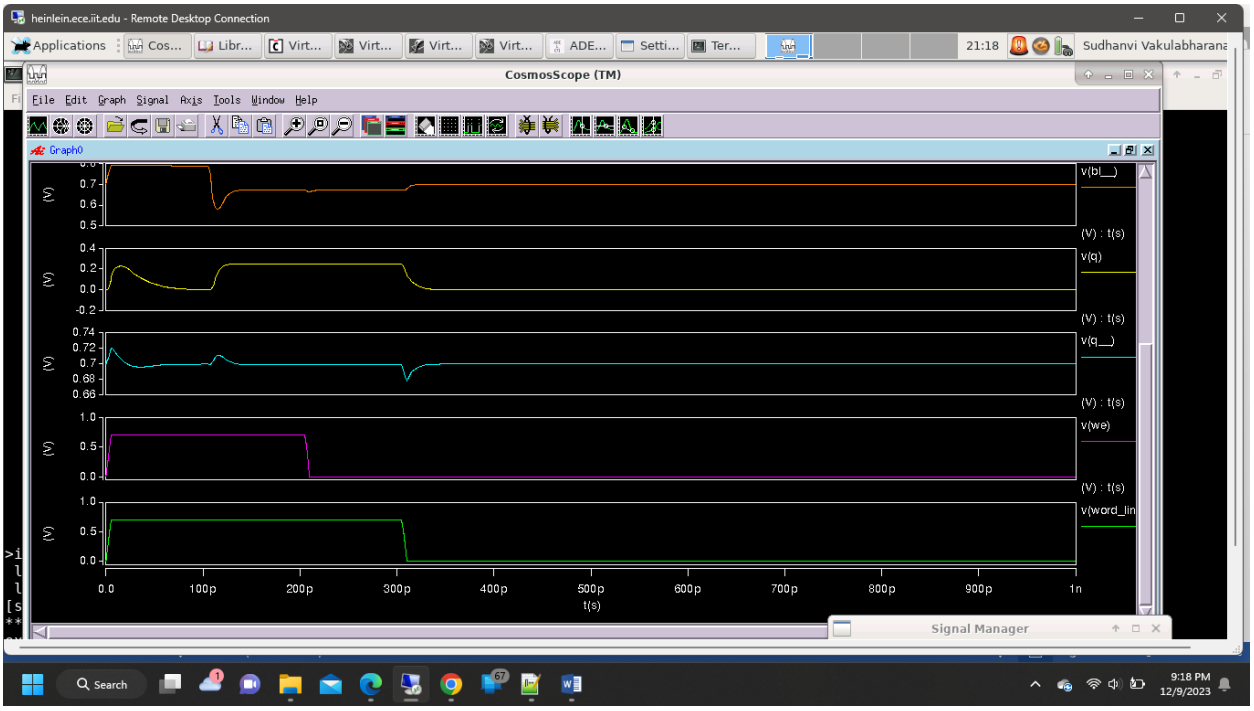


Symbol of Optimised SRAM Cell

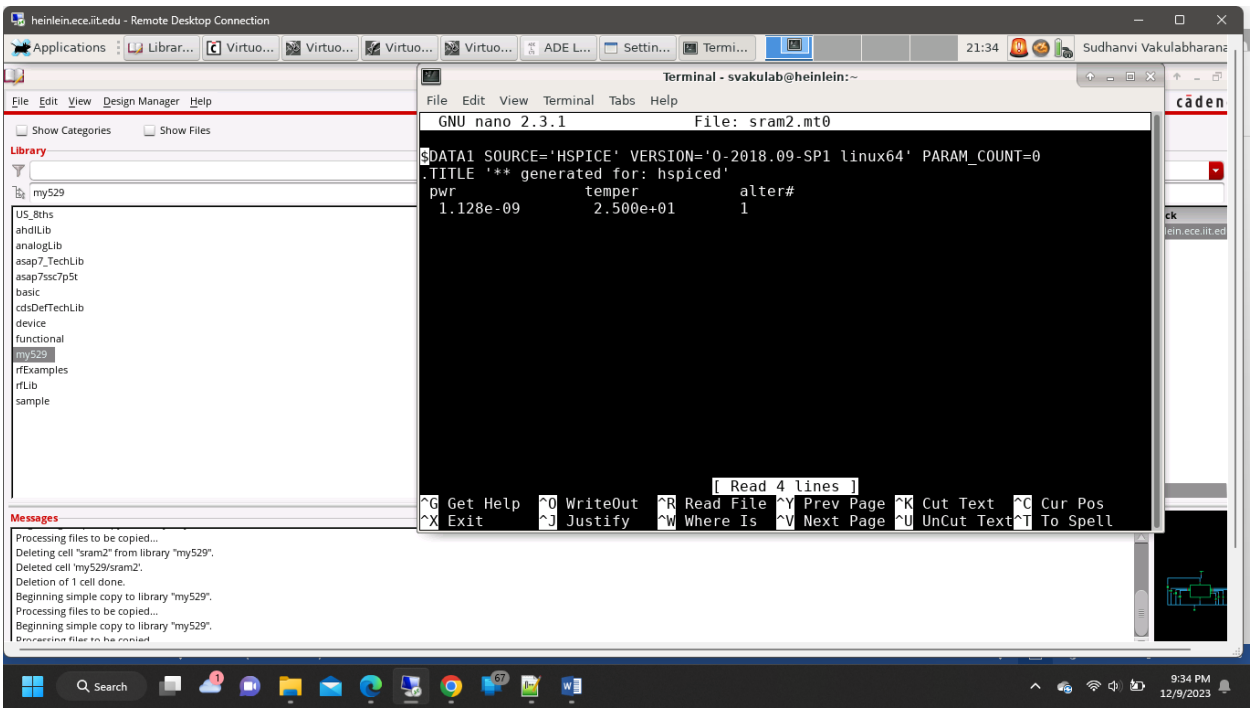


Test Schematic of Optimised 6T SRAM Cell

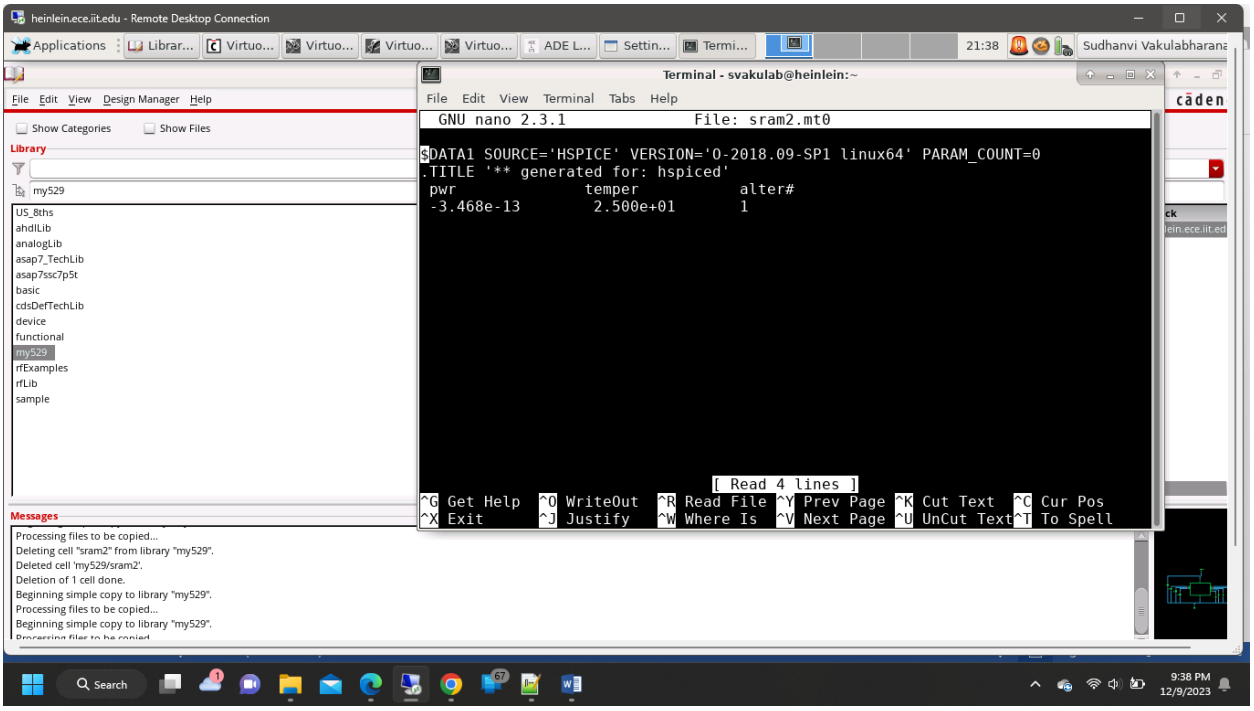




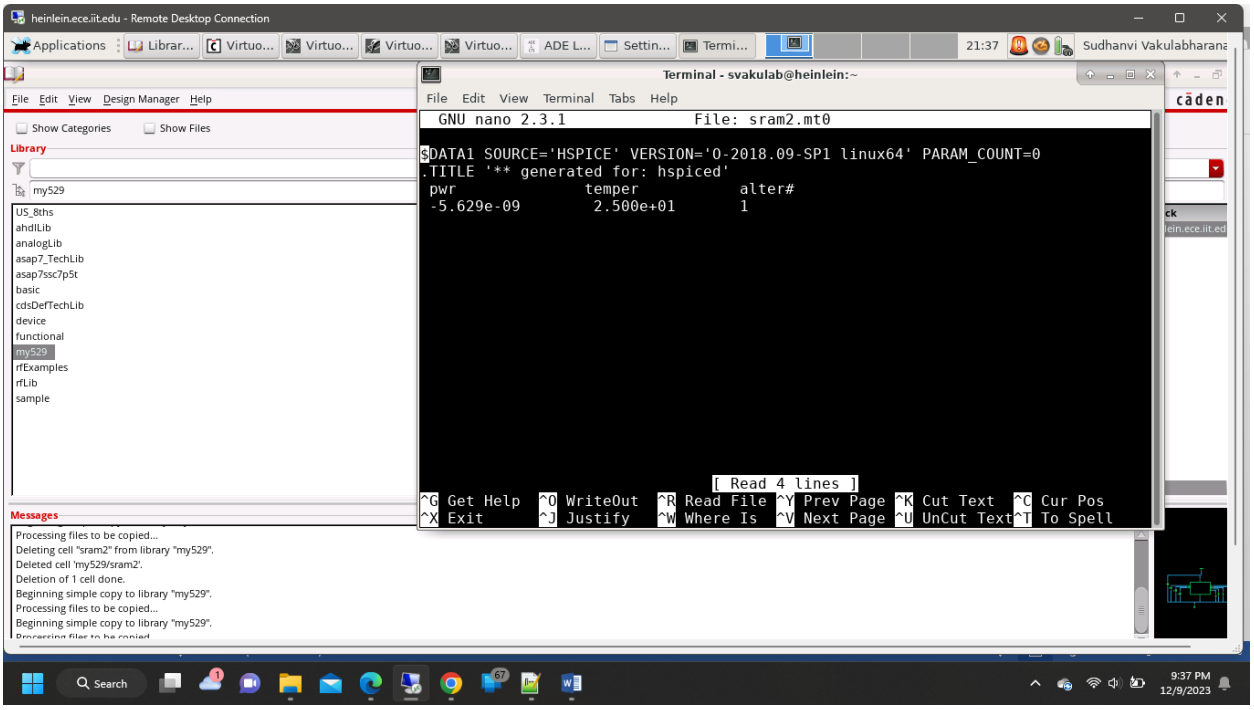
Graphs

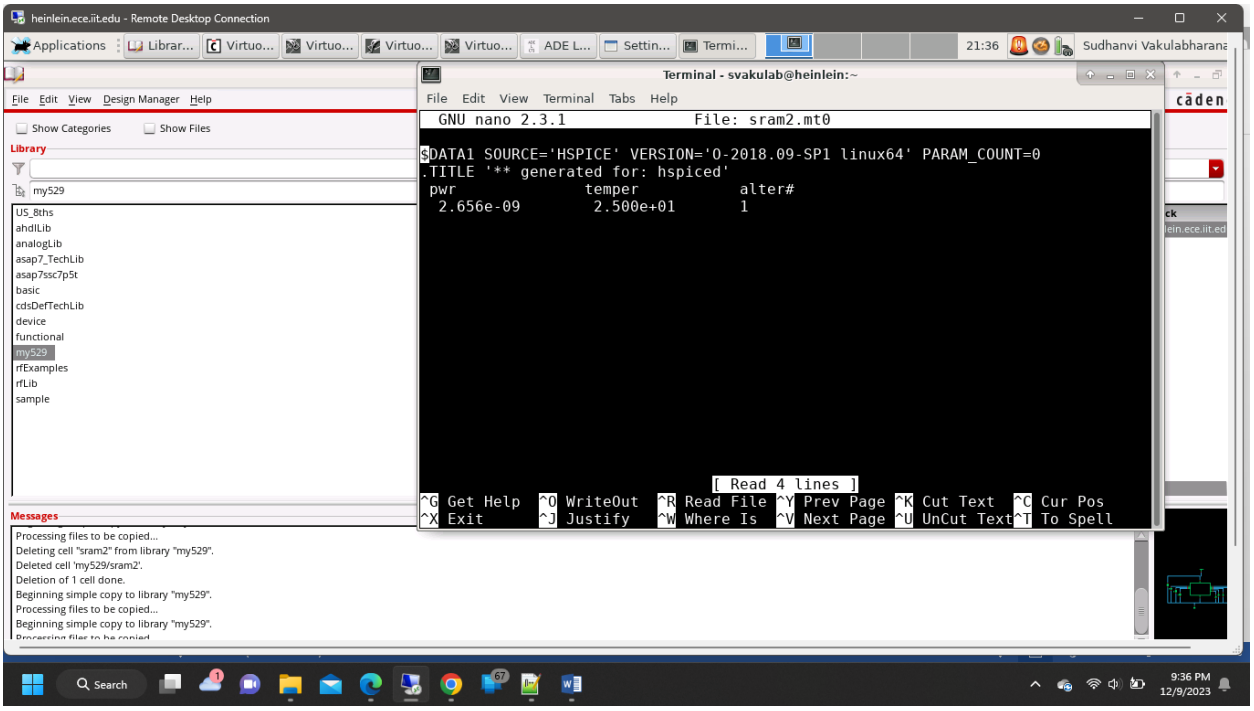


Dynamic Power at Capacitor C0

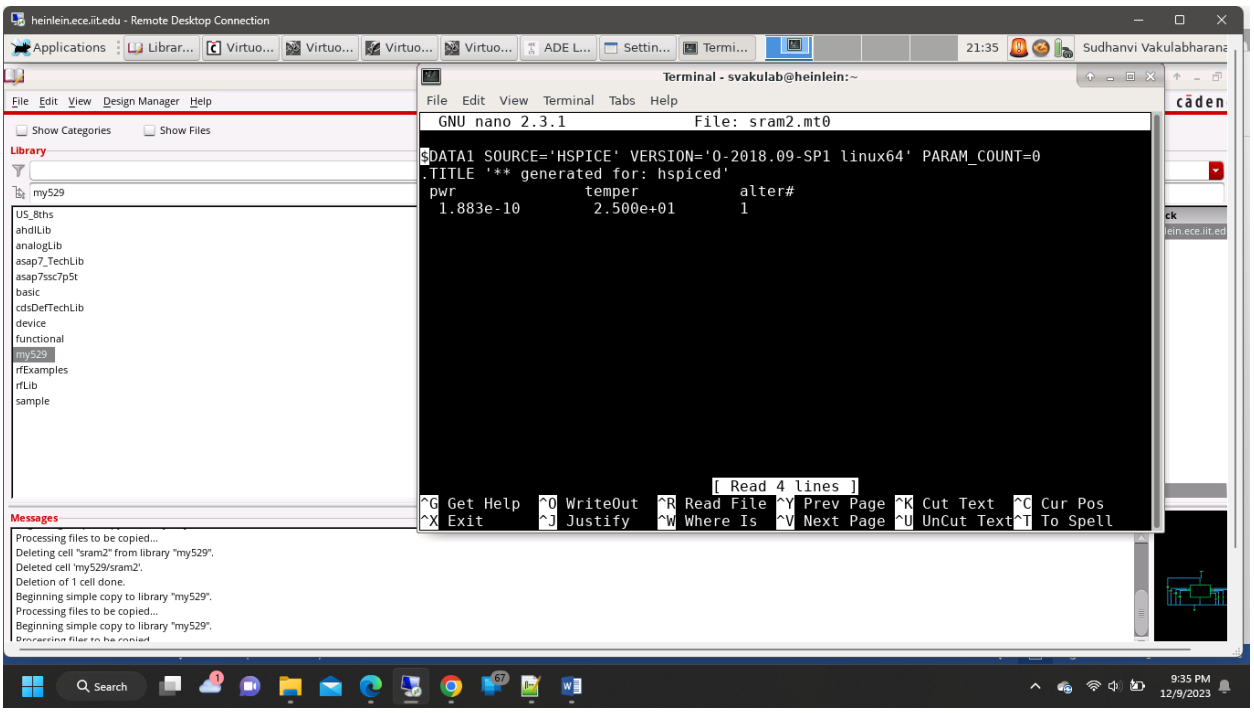


Dynamic Power at Capacitor C1





Dynamic Power at Capacitor C3



Dynamic Power at Capacitor C4

Conclusion:

In the Optimised 6T SRAM Cell we can observe that the power dissipation got reduced and this is the advantage of using the MTCMOS functionality and also by adding the additional pull down transistor.

APPENDIX

Netlist for Optimised 6T RAM Cell:

```
** Generated for: hspiceD
** Generated on: Dec  9 21:10:52 2023
** Design library name: my529
** Design cell name: sram2_test
** Design view name: schematic
.GLOBAL vdd!


.PROBE TRAN
+    V(d_o)
+    V(q__)
+    V(q)
+    V(bl__)
+    V(bl)
+    V(word_line)
+    V(we)
+    V(clk)
.TRAN 1e-12 1e-9 START=0.0

.TEMP 25.0
.OPTION
+    ARTIST=2
+    INGOLD=2
+    PARHIER=LOCAL
+    PSF=2
+    POST
.INCLUDE
"/mnt/data0/ASAP7_PDKandLIB_v1p6/asap7PDK_r1p6/models/hspice/7nm_TT_160803.pm"

** Library name: my529
** Cell name: sram1
** View name: schematic
.subckt sram1 bl bl__ clk data_out gnd q q__ vdd we word_line
M20 net13 net12 net51 vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M19 data_out net25 net51 vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M18 net25 net21 net51 vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M17 net51 net21 net21 vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M4 q q__ vdd vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M3 vdd q q__ vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M2 bl__ clk bl vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M1 bl clk vdd vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M0 vdd clk bl__ vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M16 data_out net25 gnd gnd nmos_rvt w=27e-9 l=20e-9 nfin=1
```

```
M15 gnd bl net25 gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M14 net21 bl__ gnd gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M13 net13 net12 gnd gnd nmos_rvt w=27e-9 l=20e-9 nfin=1
M12 net34 net13 net29 gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M11 net29 net12 net54 gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M10 bl we net34 gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M9 bl__ we net54 gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M8 bl word_line q gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M7 q__ word_line bl__ gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M6 q q__ gnd gnd nmos_rvt w=27e-9 l=20e-9 nfin=1
M5 gnd q q__ gnd nmos_rvt w=27e-9 l=20e-9 nfin=1
.ends sram1
** End of subcircuit definition.

** Library name: my529
** Cell name: sram2_test
** View name: schematic
xi4 bl bl__ clk d_o 0 q q__ vdd! we word_line sram1
v0 vdd! 0 DC=700e-3
v3 word_line 0 PULSE 0 700e-3 0 5e-12 5e-12 300e-12 1e-9
v2 we 0 PULSE 0 700e-3 0 5e-12 5e-12 200e-12 1e-9
v1 clk 0 PULSE 0 700e-3 0 5e-12 5e-12 100e-12 1e-9
c4 d_o 0 1e-15
c3 bl 0 1e-15
c2 bl__ 0 1e-15
c1 q 0 1e-15
c0 q__ 0 1e-15
.measure pwr AVG P(c4) FROM=0 TO=1n
.measure pwr AVG P(c3) FROM=0 TO=1n
.measure pwr AVG P(c2) FROM=0 TO=1n
.measure pwr AVG P(c1) FROM=0 TO=1n
.measure pwr AVG P(c0) FROM=0 TO=1n
.END
```

Netlist for Normal 6T SRAM cell

```
** Generated for: hspiceD
** Generated on: Dec  9 16:56:15 2023
** Design library name: my529
** Design cell name: sram1_test
** Design view name: schematic
.GLOBAL vdd!


.PROBE TRAN
+    V(d_o)
```

```
+    V(q__)
+    V(q)
+    V(bl__)
+    V(bl)
+    V(word_line)
+    V(we)
+    V(clk)
.TRAN 1e-12 1e-9 START=0.0

.TEMP 25.0
.OPTION
+    ARTIST=2
+    INGOLD=2
+    PARHIER=LOCAL
+    PSF=2
+    POST
.INCLUDE
"/mnt/data0/ASAP7_PDKandLIB_v1p6/asap7PDK_r1p6/models/hspice/7nm_TT_160803.pm"

** Library name: my529
** Cell name: sram1
** View name: schematic
.subckt sram1 bl bl__ clk data_out gnd q q__ vdd we word_line
M20 net13 net12 net51 vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M19 data_out net25 net51 vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M18 net25 net21 net51 vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M17 net51 net21 net21 vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M4 q q__ vdd vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M3 vdd q q__ vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M2 bl__ clk bl vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M1 bl clk vdd vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M0 vdd clk bl__ vdd pmos_rvt w=81e-9 l=20e-9 nfin=3
M16 data_out net25 gnd gnd nmos_rvt w=27e-9 l=20e-9 nfin=1
M15 gnd bl net25 gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M14 net21 bl__ gnd gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M13 net13 net12 gnd gnd nmos_rvt w=27e-9 l=20e-9 nfin=1
M12 net34 net13 net29 gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M11 net29 net12 net54 gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M10 bl we net34 gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M9 bl__ we net54 gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M8 bl word_line q gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M7 q__ word_line bl__ gnd nmos_rvt w=81e-9 l=20e-9 nfin=3
M6 q q__ gnd gnd nmos_rvt w=27e-9 l=20e-9 nfin=1
M5 gnd q q__ gnd nmos_rvt w=27e-9 l=20e-9 nfin=1
.ends sram1
** End of subcircuit definition.
```

```
** Library name: my529
** Cell name: sram1_test
** View name: schematic
xi4 bl bl__ clk d_o 0 q q__ vdd! we word_line sram1
v0 vdd! 0 DC=700e-3
v3 word_line 0 PULSE 0 700e-3 0 5e-12 5e-12 300e-12 1e-9
v2 we 0 PULSE 0 700e-3 0 5e-12 5e-12 400e-12 1e-9
v1 clk 0 PULSE 0 700e-3 0 5e-12 5e-12 200e-12 1e-9
c4 d_o 0 1e-15
c3 bl 0 1e-15
c2 bl__ 0 1e-15
c1 q 0 1e-15
c0 q__ 0 1e-15
.measure pwr AVG P(c4) FROM=0 TO=1n
.measure pwr AVG P(c3) FROM=0 TO=1n
.measure pwr AVG P(c2) FROM=0 TO=1n
.measure pwr AVG P(c1) FROM=0 TO=1n
.measure pwr AVG P(c0) FROM=0 TO=1n
.END
```

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